Lecture 4: PRG *Constructions*

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> ***Warning:*** This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Recap

Last lecture we covered the following:

1. We defined the notion of a pseudorandom generator.

   Definition 1. An efficient (poly-time computable) deterministic function $G : \{0,1\}^\lambda \to \{0,1\}^{n(\lambda)}$ is said to be a pseudorandom generator if the following two conditions hold:

   - It is expanding; i.e., $n(\lambda) > \lambda$.
   - It is pseudorandom, i.e.

     $$\{G(U_\lambda)\}_{\lambda\in\mathbb{N}} \approx \{U_{n(\lambda)}\}_{\lambda\in\mathbb{N}},$$

     where U_ℓ is the uniform distribution over $\{0,1\}^\ell$.

2. We proved that pseudorandomenss is equivalent to the following "next-bit unpredictability" property.

   Definition 2. An expanding function $G : \{0,1\}^\lambda \to \{0,1\}^{n(\lambda)}$ is next-bit unpredictable if for every poly-size adversary $\mathcal{A}$ there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$ and every $i \in [n(\lambda)]$

   $$\Pr_{U\leftarrow\{0,1\}^\lambda}[\mathcal{A}(G(U_\lambda)_{[i-1]}) = G(U_\lambda)_i] = 1/2 + \mu(\lambda)$$

   where $G(U_\lambda)_{[i-1]}$ denotes the first $i-1$ bits of $G(U_\lambda)$ and $G(U_\lambda)_i$ denotes the i'th bit of $G(U_\lambda)$.

3. We proved that if there exists a PRG $G : \{0,1\}^\lambda \to \{0,1\}^{n(\lambda)}$ then there exists a computationally secure encryption scheme with key space $\mathcal{K}_\lambda = \{0,1\}^\lambda$ and message space $\mathcal{M}_\lambda = \{0,1\}^{n(\lambda)}$, defined by

   $$\mathsf{Enc}(k,m) = G(k) \oplus m$$

   and

   $$\mathsf{Dec}(k,c) = G(k) \oplus c.$$

Today:

In this lecture (and next), our goal is to construct a PRG

$$G : \{0,1\}^{\lambda} \to \{0,1\}^{n(\lambda)}$$

for an arbitrary polynomial expanding function $n : \mathbb{N} \to \mathbb{N}$.

We will focus on constructing a PRG G that expands by a single bit; i.e,

$$G : \{0,1\}^{\lambda} \to \{0,1\}^{\lambda+1}$$

This is enough since we can generically increase the stretch of a PRG.

Stretching a PRG

One can increase the stretch of a PRG by applying a PRG to the output of the PRG. Namely, given a PRG

$$G : \{0,1\}^{\lambda} \to \{0,1\}^{\lambda+1}$$

one can construct a PRG

$$G^k : \{0,1\}^{\lambda} \to \{0,1\}^{\lambda+k}$$

where

$$G^k(U_\lambda) = G(G(\ldots G(U_\lambda)\ldots)).$$

Theorem 3. *[2] If G is a* PRG *then for every polynomial $k = k(\lambda)$, G^k is a* PRG.

Proof. Fix a polynomial $k = k(\lambda) \geq 1$. We need to prove that G^k is a PRG. The fact that G^k is stretching and efficiently computable follows from the fact that G satisfies these properties, together with the fact that $k(\lambda) \leq \mathrm{poly}(\lambda)$. The proof that it is pseudorandom is by induction on k. Intuitively, the proof proceeds as follows:

Base case: $k = 1$. By assumption

Induction step: Suppose $G^{k-1} : \{0,1\}^{\lambda} \to \{0,1\}^{\lambda+k-1}$ is pseudorandom, and we will prove that $G^k : \{0,1\}^{\lambda} \to \{0,1\}^{\lambda+k}$ is pseudorandom. Fix a poly-size adversary $\mathcal{A}$. Then for every $\lambda \in \mathbb{N}$

$$\begin{aligned}
&|\Pr[\mathcal{A}(G^k(U_\lambda)) = 1] - \Pr[\mathcal{A}(U_{\lambda+k}) = 1]| = \\
&|\Pr[\mathcal{A}[G(G^{k-1}(U_\lambda)) = 1] - \Pr[\mathcal{A}(U_{\lambda+k}) = 1]| = \\
&|\Pr[\mathcal{A}[G(G^{k-1}(U_\lambda)) = 1] - \Pr[\mathcal{A}[G(U_{\lambda+k-1}) = 1] + \Pr[\mathcal{A}[G(U_{\lambda+k-1}) = 1] - \Pr[\mathcal{A}(U_{\lambda+k}) = 1]| \leq \\
&|\Pr[\mathcal{A}[G(G^{k-1}(U_\lambda)) = 1] - \Pr[\mathcal{A}[G(U_{\lambda+k-1}) = 1]| + |\Pr[\mathcal{A}[G(U_{\lambda+k-1}) = 1] - \Pr[\mathcal{A}(U_{\lambda+k}) = 1]| \triangleq \\
&\mu_{k-1}(\lambda) + \mu_1(\lambda).
\end{aligned}$$

Note that μ_{k-1} is a negligible function by the induction hypothesis, and μ_1 is a negligible function (for every $k \leq \text{poly}(\lambda)$) by assumption that G is a PRG, and the sum of negligible functions is negligible.

Remark. It is important that the blowup is *additive*, and each time we add a negligible function μ_1, which is negligible for every $k \leq \text{poly}(\lambda)$. If say, μ_1 was replaced by μ_{k-1} then each time the negligible function would increase by a multiplicative factor of 2, and thus we could have applied the induction only $k = O(\log \lambda)$ many times.

□

The above theorem implies that it suffices to construct a PRG that has a single bit stretch. The following remarkable theorem is known.

Theorem 4. *[2] Pseudorandom generators exist assuming the existence of one-way functions.*

This theorem has a beautiful but very complicated proof. We will prove a simplified version that assumes the existence of one-way permutations, defined below.

Definition 5. A function $f : \{0,1\}^* \to \{0,1\}^*$ is a permutation if it is length preserving and bijective; namely, for every $\lambda \in \mathbb{N}$ and for every $x \in \{0,1\}^\lambda$ it holds that $f(x) \in \{0,1\}^\lambda$, and for every $y \in \{0,1\}^\lambda$ there is a unique $x \in \{0,1\}^\lambda$ such that $f(x) = y$.

Definition 6. A one-way permutation (OWP) $f : \{0,1\}^* \to \{0,1\}^*$ is a one-way function that is also a permutation.

We prove the following theorem.

Theorem 7. *Pseudorandom generators exist assuming the existence of a* OWP*.*

By Theorem 3, to prove Theorem 7 it suffices to construct a PRG that stretches by a single bit assuming the existence of a OWP.

PRG *Construction with a Single Bit Stretch*

Let $f : \{0,1\}^* \to \{0,1\}^*$ be a one-way permutation. Suppose that f has a *hardcore predicate* $P : \{0,1\}^* \to \{0,1\}$.

Definition 8. $P : \{0,1\}^* \to \{0,1\}$ is a *hardcore predicate* of $f : \{0,1\}^* \to \{0,1\}^*$ if it is efficiently computable and for every poly-size $\mathcal{A}$ there exists a negligible function $\mu : \mathbb{N} \to [0,1]$ such that for every $\lambda \in \mathbb{N}$,

$$\Pr[\mathcal{A}(f(U_\lambda)) = P(U_\lambda)] \leq \frac{1}{2} + \mu(\lambda).$$

Given a one-way permutation f with a hardcore predicate P, let

$$G(U_\lambda) = f(U_\lambda) \circ P(U_\lambda)$$

Theorem 9. *G is a pseudorandom generator.*

Proof. G is expanding by definition, and the fact that it is efficiently computable follows from the fact that both f and P are efficiently computable. Thus, it remains to prove that G is pseudorandom, or equivalently that G satisfies the next-bit unpredictability property.

To this end fix any poly-size adversary $\mathcal{A}$. We need to prove that there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$ and every $i \in [\lambda + 1]$

$$\Pr[\mathcal{A}(G(U)_{[i-1]}) = G(U)_i] \leq \frac{1}{2} + \mu(\lambda)$$

The fact that f is a one-way permutation implies that for every $i \leq [\lambda]$

$$\Pr[\mathcal{A}(G(U)_{[i-1]}) = G(U)_i] = \frac{1}{2}.$$

For $i = \lambda + 1$, the fact that P is a hardcore predicate of f implies that there exists a negligible function μ such that

$$\Pr[\mathcal{A}(G(U)_{[\lambda]}) = G(U)_{\lambda+1}] = \Pr[\mathcal{A}(f(U_\lambda)) = P(U_\lambda)] \leq \frac{1}{2} + \mu(\lambda),$$

as desired. ☐

In theorem 7 we assumed there exists a one-way permutation but we did not assume that it has a hardcore predicate. Thankfully, Goldreich and Levin proved that every one-way function has a hardcore predicate!

Theorem 10. *[1] If f is a one-way function, then the following randomized predicate*

$$P(x, r) := x \cdot r \mod 2 = \sum_{i \in \lambda} x_i r_i \mod 2$$

is a hardcore predicate for f. Namely, for every poly-size $\mathcal{A}$ there exists a negligible function $\mu : \mathbb{N} \to [0, 1]$ such that for every $\lambda \in \mathbb{N}$

$$\Pr_{U_\lambda \leftarrow \{0,1\}^\lambda}[\mathcal{A}(f(U_\lambda), r) = P(U_\lambda, r))] \leq \frac{1}{2} + \mu(\lambda)$$

Proof. Suppose for contradiction that there exists a poly-size adversary $\mathcal{A}$ and a non-negligible function $\epsilon : \mathbb{N} \to [0, 1]$ such that for every $\lambda \in \mathbb{N}$,

$$\Pr_{U_\lambda \leftarrow \{0,1\}^\lambda}[\mathcal{A}(f(U_\lambda), r) = P(U_\lambda, r))] \geq \frac{1}{2} + \epsilon(\lambda)$$

We will show how to use $\mathcal{A}$ to construct an adversary $\mathcal{B}$ that breaks the one-wayness of f.

Since $\mathcal{A}$ is only assumed to predict with small advantage ϵ, we cannot expect $\mathcal{B}$ to invert every given image $y = f(x)$. Indeed, it may be that $\mathcal{A}$ always fails when given an input whose first λ-bits is y. So which y's can we hope to invert?

Define

$$\mathsf{GOOD} = \{x \in \{0,1\}^\lambda : \Pr[\mathcal{A}(f(x), r) = x \cdot r] \geq \frac{1}{2} + \frac{\epsilon(\lambda)}{2}$$

By Markov's inequality

$$\Pr_{x \leftarrow U_\lambda}[x \in \mathsf{GOOD}] \geq \frac{\epsilon(\lambda)}{2}$$

To expand on this, denoting by

$$p = \Pr_{x \leftarrow U_\lambda}[x \in \mathsf{GOOD}],$$

it holds that

$$\begin{aligned}
&\frac{1}{2} + \epsilon(\lambda) \leq \\
&\Pr_{x \leftarrow U_\lambda}[\mathcal{A}(f(x), r) = P(x, r)] = \\
&\Pr_{x \leftarrow \mathsf{GOOD}}[\mathcal{A}(f(x), r) = P(x, r)] \cdot \Pr_{x \leftarrow U_\lambda}[x \in \mathsf{GOOD}] + \Pr_{x \leftarrow \neg\mathsf{GOOD}}[\mathcal{A}(f(x), r) = P(x, r)] \cdot \Pr_{x \leftarrow U_\lambda}[x \notin \mathsf{GOOD}] = \\
&\Pr_{x \leftarrow \mathsf{GOOD}}[\mathcal{A}(f(x), r) = P(x, r)] \cdot p + \Pr_{x \leftarrow \neg\mathsf{GOOD}}[\mathcal{A}(f(x), r) = P(x, r)] \cdot (1 - p) \leq \\
&p + \frac{1}{2} + \frac{\epsilon(\lambda)}{2},
\end{aligned}$$

which implies that indeed $p \geq \frac{\epsilon(\lambda)}{2}$.

We will construct an algorithm $\mathcal{B}$ that succeeds in inverting every $f(x)$ for $x \in \mathsf{GOOD}$ with non-negligible probability.

Special case 1: $\mathcal{A}$ predicts $P(x, r)$ perfectly for every $x \in \mathsf{GOOD}$

As a warmup, suppose that $\mathcal{A}$ is a perfect predictor that for every $x \in \mathsf{GOOD}$ and every $r \in \{0,1\}^\lambda$

$$\mathcal{A}(f(x), r) = x \cdot r$$

With such a perfect predictor it is easy to invert the OWF f. Specifically, the inverter $\mathcal{B}$, given $y = f(x)$ where $x \leftarrow U_\lambda$, does the following:

1. For every $i \in [\lambda]$ compute $x'_i = \mathcal{A}(y, e_i)$.
2. Output $x = (x'_1, \ldots, x'_\lambda)$.

By our assumption that $\mathcal{A}$ is a perfect predictor, it follows that

$$x'_i = x \cdot e_i = x_i,$$

as desired.

Unfortunately, our predictor $\mathcal{A}$ may not be perfect. Next, we relax the perfect condition as follows.

Special case 2: There exists a non-negligible $\epsilon = \epsilon(\lambda)$ such that for every $x \in \mathsf{GOOD}$, $\mathcal{A}$ predicts $P(x, r)$ with probability $3/4 + \epsilon$

We can use this predictor to construct an algorithm $\mathcal{B}$ that for every $x \in \mathsf{GOOD}$, given $f(x)$ finds an inverse with overwhelming probability (i.e., probability $1 - \mu$ for some negligible function μ). The inverter $\mathcal{B}$, given $y = f(x)$ where $x \leftarrow U_\lambda$, does the following:

1. Set $k = \lambda/\epsilon^2$.
2. For every $i \in [\lambda]$, do the following:
   (a) Choose at random $r_{i,1}, \dots, r_{i,k} \leftarrow \{0,1\}^\lambda$.
   (b) For every $j \in [k]$ compute $b_{i,j} = \mathcal{A}(y, r_{i,j})$ and $b'_{i,j} = \mathcal{A}(y, r_{i,j} \oplus e_i)$, and let $x_{i,j} = b_{i,j} \oplus b'_{i,j}$.
   (c) Let $x'_i = \mathsf{majority}\{x_{i,1} \dots, x_{i,k}\}$
3. Output $x' = (x'_1, \dots, x'_\lambda)$

Note that by our assumption for every $i \in [\lambda]$ and $j \in [k]$,

$$\Pr_{r_{i,j} \leftarrow \{0,1\}^\lambda}[x'_{i,j} = x_i] \geq$$

$$\Pr_{r_{i,j} \leftarrow \{0,1\}^\lambda}[\mathcal{A}(y, r_{i,j}) = x \cdot r_{i,j} \quad \wedge \quad \mathcal{A}(y, r_{i,j} \oplus e_i) = x \cdot (r_{i,j} \oplus e_i)] =$$

$$1 - \Pr_{r_{i,j} \leftarrow \{0,1\}^\lambda}[\mathcal{A}(y, r_{i,j}) \neq x \cdot r_{i,j} \quad \vee \quad \mathcal{A}(y, r_{i,j} \oplus e_i) \neq x \cdot (r_{i,j} \oplus e_i)] \geq$$

$$1 - \Pr_{r_{i,j} \leftarrow \{0,1\}^\lambda}[\mathcal{A}(y, r_{i,j}) \neq x \cdot r_{i,j}] - \Pr_{r_{i,j} \leftarrow \{0,1\}^\lambda}[\mathcal{A}(y, r_{i,j} \oplus e_i) \neq x \cdot (r_{i,j} \oplus e_i)] \geq$$

$$1 - 1/2 + 2\epsilon = 1/2 + 2\epsilon$$

Note that for every $i \in [\lambda]$ it holds that $x'_{i,1}, \dots, x'_{i,k}$ are independent Boolean random variables such that $\Pr[x'_{i,j} = x_i] \geq 1/2 + 2\epsilon$. Therefore, by the Chernoff bound

$$\Pr[\mathsf{majority}\{x_{i,1}, \dots, x_{i,\lambda}\} \neq x_i] \leq 2^{-O(\epsilon^2 \cdot k)}$$

Theorem 11 (Chernoff bound). *Let $X_1, X_2, \dots, X_n$ be independent random variables taking values in $\{0,1\}$ (Bernoulli trials). Let*

$$X = \sum_{i=1}^{n} X_i, \qquad \mu = \mathbb{E}[X].$$

For any $0 < \delta < 1$*:*

$$\Pr[X \geq (1+\delta)\mu] \leq \exp\left(-\frac{\delta^2}{3}\mu\right),$$

and

$$\Pr[X \leq (1-\delta)\mu] \leq \exp\left(-\frac{\delta^2}{2}\mu\right).$$

Thus, for every $i \in [\lambda]$

$$\Pr[\mathcal{B}(f(x)) = x' : \ x'_i \neq x_i] \leq 2^{-O(\lambda)},$$

which together with the union bound implies that

$$\Pr[\mathcal{B}(f(x)) \neq x] \leq \lambda \cdot 2^{-O(\lambda)} = \mathrm{negl}(\lambda).$$

Remark. Note that we could have chosen the same random $r_1, \ldots, r_k \leftarrow \{0,1\}^\lambda$ for all the coordinates $i \in [\lambda]$. The reason is that we do not need independence for the union bound.

General case: There exists a non-negligible $\epsilon = \epsilon(\lambda)$ *such that for every* $x \in \mathsf{GOOD}$*,* $\mathcal{A}$ *predicts* $P(x,r)$ *with probability* $1/2 + \epsilon$

Note that in the above analysis the reason that we needed a success probability of greater than $\frac{3}{4}$ is because, in order to get a non-negligible probability of guessing a single bit x_i we needed to combine *two* guesses for two bits $x \cdot r_{i,j}$ and $x \cdot (r_{i,j} \oplus e_i)$.

The Goldreich-Levin reduction starts with the following (ridiculous!) idea. Suppose that for each pair r and $r \oplus e_i$, we simply guess the bit $x \cdot r$ ourselves, and we only use the adversary $\mathcal{A}$ to guess the bit $x \cdot (r \oplus e_i)$. If we could somehow manage to always guess the bit $x \cdot r$ correctly then we could use the same argument as above since under this (ridiculous) assumption both bits $x \cdot r$ and $x \cdot (r \oplus e_i)$ are guessed correctly with probability at least $\frac{1}{2} + \epsilon$.

The problem is that we cannot guess $x \cdot r$ by ourselves with probability better than random. This is exactly the task that we needed $\mathcal{A}$ to do in the first place! But we can guess each of these bits at random and have a success probability $\frac{1}{2}$. This seems useless since in the above procedure, for each $i \in [\lambda]$ we needed to choose a total of k different values of $x \cdot r_j$, and if we guess each of them at random then we will guess all of them correctly only with probability 2^{-k}. But it turns out that this random guessing is actually really useful, due to the following clever idea: suppose that we happen to correctly guess $x \cdot s_1$ and $x \cdot s_2$. Then we can also guess correctly

$$x \cdot (s_1 \oplus s_2) = (x \cdot s_1) \oplus (x \cdot s_2)$$

More generally, suppose we guessed correctly

$$x \cdot r_1, \ x \cdot r_2, \ \ldots, \ x \cdot r_\ell$$

then we can use these ℓ bits to guess correctly

$$x \cdot (\oplus_{i \in I} r_i) = \oplus_{i \in I}(x \cdot s_i)$$

for every $I \subseteq [\ell]$. Since there are 2^ℓ such subsets we learned 2^ℓ inner products $x \cdot r_I$, where $r_I = \oplus_{i \in I} r_i$. Therefore, in order to learn k different inner products $x \cdot r_I$ we need to guess only $\ell = \log k$ many bits of the form $x \cdot r_j$! And we can guess all ℓ inner products correctly with probability $\frac{1}{k}$.

You may be concerned that now all these r_I's are not independent. Indeed, they are not! For example, $r_{\{1,2\}} = r_1 \oplus r_2$. As a result we cannot use the Chernoff bound in the analysis. However, they are *pairwise independent*. Therefore, instead of Chernoff bound we can rely on Chebyshev's inequality which is a weaker concentration bound.

Theorem 12 (Chebyshev's inequality)**.** *Let $Z_1, \ldots, Z_m$ be pairwise independent random variables obtaining values in $\{0,1\}$, such that for every $j \in [m]$ $\Pr[Z_j = 1] = p$. Then, for every $\delta > 0$,*

$$\Pr\left[\left|\frac{\sum_{j=1}^m Z_j}{m} - p\right| \geq \delta\right] \leq \frac{1}{4\delta^2 m}$$

So, the inverter $\mathcal{B}$, given $y = f(x)$, does the following:

1. Set $k = \lambda/\epsilon^2$.
2. For every $i \in [\lambda]$, do the following:
   (a) Set $\ell \approx \log k$
   (b) choose at random $r_1, \ldots, r_\ell \leftarrow \{0,1\}^\lambda$.
   (c) For every $j \in [\ell]$, choose at random $b_j \leftarrow \{0,1\}$.[1]
   (d) For every $J \subseteq [\ell]$, let
   $$r_J = \oplus_{j \in J} r_j \quad \text{and} \quad b_J = \oplus_{j \in J} b_j.$$
   (e) Similarly, for every $J \subseteq [\ell]$, let
   $$b'_{i,J} = \mathcal{A}(y, r_J \oplus e_i) \quad \text{and} \quad x_{i,J} = b_J \oplus b'_{i,J}.$$
   (f) Let $x'_i = \mathsf{majority}\{x_{i,J}\}_{J \subseteq [\ell]}$
3. Output $x' = (x'_1, \ldots, x'_\lambda)$

[1] b_j is a guess for $x \cdot r_j$.

Suppose that all our ℓ guesses where correct, which happens with probability $2^{-\ell} = \text{poly}(k)$. In this case, as above for every $i \in [\lambda]$ and

$J \in [\ell]$,

$$\Pr_{r_1,\ldots,r_\ell \leftarrow \{0,1\}^\lambda}[x'_{i,J} = x_i] \geq$$
$$\Pr_{r_1,\ldots,r_\ell \leftarrow \{0,1\}^\lambda}[\mathcal{A}(y,(r_J + e_i)) = x \cdot (r_J \oplus e_i)] =$$
$$1 - \Pr_{r_1,\ldots,r_\ell \leftarrow \{0,1\}^\lambda}[\mathcal{A}(y, r_J \oplus e_i) \neq x \cdot (r_J \oplus e_i)] \geq$$
$$1/2 + \epsilon.$$

This seems great, except that now we cannot apply the Chernoff bound since the random variables $\{r_J\}_{J \subseteq [\ell]}$ are not independent! However, as mentioned above they are pairwise independent, and hence we can rely on Chebyshev's inequality to argue that for every $x \in \mathsf{GOOD}$ and every $i \in [\lambda]$,

$$\Pr[x'_i \neq x_i] \leq \frac{1}{4\epsilon^2 k}$$

In particular, setting $k = \lambda \cdot \epsilon^{-2}$, we can take a union bound, and guarantee that for every $x \in \mathsf{GOOD}$,

$$\Pr[\mathcal{B}(f(x) = x] \geq \frac{1}{4},$$

contradicting the fact that f is a one-way function. $\square$

References

[1] Oded Goldreich and Leonid A. Levin. A hard-core predicate for all one-way functions. In *Proceedings of the 21st Annual ACM Symposium on Theory of Computing (STOC)*, pages 25–32, Seattle, Washington, USA, 1989.

[2] Johan Håstad, Russell Impagliazzo, Leonid A. Levin, and Michael Luby. A pseudorandom generator from any one-way function. *SIAM Journal on Computing*, 28(4):1364–1396, 1999.